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Selective Growth Mechanism (SGM). Powder bed 3D printing with mycelium-based materials

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ABSTRACT: This paper presents a novel approach to 3D print heterogeneous, reusable 3D-printed structures that integrate the morphological characteristics of natural mycelium growth combined with precision injection 3D printing, enabling the creation of objects with varied and tailored properties. Current 3D printing methods involving mycelium are primarily constrained to extruding materials with homogeneous properties, leading to concentrated growth along outer boundaries. In contrast, the proposed method facilitates controlled growth and incorporates monitoring chamber equipped with sensors and a camera, functioning similarly to a bioreactor, that tracks mycelial behaviour in real time. This technique advances the understanding of mycelial growth dynamics and the parameters influencing it, offering a foundation for the design of increasingly complex, biologically derived components in the future.

1 INTRODUCTION

The application of 3D printing to mycelium materials is gaining global interest in design and engineering. This research, evolving alongside molding and subtractive methods, enhances flexibility, sustainability, and complexity in fabrication. Both 3D printing and the use of living organisms reveal intricate behaviors, introducing biological morphological features. This article presents a semi-systematic review of studies on 3D printing with Lignicolous mycelium growth, comparing them within the material design space. From this review, we identify a research gap as across the state-of-the-art. In an effort to increase the possibility of leveraging this unique characteristic of fabricating with filamentous fungi, we then present a novel 3D printing platform inspired by powder bed technologies which we name Selective Growth Mechanism (SGM).

This novel method leverages from mycelium's natural growth to bind organic substrates with precise dispensing of liquid in powder, blending digital and natural processes. It distinguishes from the state-of-the-art mycelium 3D printing, which often predefined geometry is inoculated before or after the production. This method leverages natural growth to shape form and material properties, integrating advanced fungal biology into material design.

Integrating a fungal organism requires active monitoring of environmental conditions, therefore our 3D printer chamber is equipped with temperature, humidity, and air quality sensors, as well as an RGB camera to monitor growth. Using a research-by-design approach, we experiment with two inoculation methods — one with chlamydospores in pre-inoculated and non-inoculated substrates — and two moisturization methods (water and mycelium solution). These experiments and insights into fungal biology outline the new design space supported by SGM. Finally, we discuss the technology's benefits, limitations, and potential research avenues.

2 CONTEXT

Filamentous fungi have seen a remarkable increase in interest with the growing popularity of amateur mycology in the past fifty years (Steinhardt 2018). A recent survey in studies reporting on

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MBC mechanical characteristics already accounted fifteen species in use and more than sixteen principal substrates (Rigobello 2023). Considering the popular use of medium modifying supplements such as brans or calcium carbonate, and the diversity of published cultivation and transformation methods reveals the uncharted knowledge's breadth in mycelium materials description. What is more, mycelium materials have recently been classified in the wider group of Engineered-Living Materials (ELMs). Researchers active in that field now envision the engineering of mycelial morphologies and masterly designed, potentially sustainable materials by leveraging microbial metabolic activities (Carlström et al. 2024). The reciprocity between the developed simulations can be implemented in 3D printing technologies to investigate and validate the outcomes.

Lignicolous filamentous fungi can be cultivated using various methods. These methods utilise principles of either Solid-State Fermentation (SSF) - consisting in cultivating microbes on an agar medium or on lignocellulosic particles here - or Liquid-State Fermentation (LSF)—used either statically or in submerged, agitated state. In the large majority of cases in the literature, MBC are being described as homogeneous materials (Islam et al. 2018). As the thermodynamics and fluid mechanics involved in MBCs' SSF may hinder the attaining of homogeneous material properties because white rot fungi rely on aerobic respiration to develop biomass (Carlström et al. 2024).

The design space supported by these strategies talks to millimeter and centimeter scales, with strategies consisting in composing plant material — prospectively inorganic materials too — as substrates or in-fills. In the state-of-the-art the capacity of fungi to produce differentiated morphologies, such as fruiting bodies or aerial hyphae, has not been discussed in relation to mycelium material properties to the best of our knowledge. These morphologies might be engineered to induce locally differentiated and graded material properties, varying elasticity, density, hyphal orientation, or biochemical gradients for example, and relate to cellular and molecular differentiation. These variations can be expressed in the form of productions of multicellular morphologies from units seeing little particularisation themselves, hyphae. As such, we can identify morphologies like Sclerotia, Pseudosclerotia, Rhizomorphs, and fruit bodies, which themselves can be composed of a wide variety of distinct hyphal macrostructures across species (for *Fomes fomentarius* fruit bodies, see (Pylkkänen et al. 2023))

All of these structures are composed of little diverging thin-walled generative, thick-walled skeletal, and branched binding hyphae. Fruit bodies, in all their diversity, are produced only from generative hyphae that see variation in diameter, septa spacing, and branching frequency. Although the mechanisms of environmentally-triggered differentiation for fruit bodies promotion are generally well understood for agronomic and economic interests, many of the conditions of morphological expression are not completely described yet.

To satisfy the contemporary interest for manufacturing ELMs with finely modified properties leveraging understandings of filamentous fungi metabolism, new cultivation processes are necessary. 3D printing can enable implementing the different environmental and growing factors in terms of material (Soh et al. 2023), geometry (Gomaa et al. 2024) scale (Mohseni et al. 2023), among others; however, conventional 3D printing methods with mycelium focus mainly on extruded material which limits the morphological growth of mycelium in space and results in homogeneous material properties. The present study seeks to contribute to the advances in cultivation process development, therefore the design of the 3D printer is similar to a bioreactor, to create a cultivation platform, permitting the engineering of complex MBC ELMs and the investigation of the conditions of morphological differentiations in filamentous fungi.

3 METHOD

Fungi can sense and react to environmental triggers such as intra- and extracellular nutrient levels, pheromones, pH, water, oxidative stress, temperature, light intensity and wavelength, gravity, and mechanical stress. This represents a wide set of variables to try and influence hyphal differentiation in SSF. These variables broadly classify as endogenous and exogenous to MBCs. Endogenous ones cover the substrate's chemical profile, such as organic carbon content, carbon to nitrogen ratio, ashes content, or moisture content, substrate's geometrical specifics such as particles shape, size and distribution, and finally fungal strain characteristics & inoculation method. Exogenous

variables relate to the environment of the sample to grow: temperature, light conditions, relative humidity, and gas composition.

MBCs are volumetric objects, as compared with sheet-like materials. An advance in cultivation process control would then be most useful if it were to bring forth control in three dimensions, and offer the capacity to intervene at the core of a fermenting material. While our aim in the present study is to establish a scalable method of process control that can support the cultivation and study of morphological differentiation, we focus on a first stage consisting on the description of growth control in three dimensions, for integrating more control once this established. To decide on the growth control variable to use for the prototyping, the previously scoped variables were classified in Table 1 against the cultivation stage at which they can be controlled, the necessary sterility for intervention, and resolution of control. Based on this classification, moisture content was selected because it allows to work with a relatively low level of sterility that facilitates prototyping and iteration, has a relatively high resolution of control, and can be continuously delivered during fermentation. Furthermore, water characteristics have been identified as a major predictor of fungal decay in relation to MBC (Rigobello 2023).

Certain basidiomycetes can produce chlamydospores in their mycelia which are specialised, asexual spore structures. These spores are thick-walled and particularly resistant to rough environmental conditions. They can be produced in parallel to the principal pathway for producing sexual spores which occurs in fruit bodies. Chlamydospore synthesis has been previously reported in some *Ganoderma* spp., of which *Ganoderma lucidum* (Moradali et al. 2007; Stalpers 2000; Loyd et al. 2019), or *Trametes versicolor* (Bhatt et al. 2016) for example, which are species of interest for MBC cultivation (Rigobello & Ayres 2023). We can note that chlamydospore synthesis capacity has been used for ten years now by Ecovative Design LLC to commercialise Grow-It-Yourself kits based on a *Ganoderma sessile* strain that ship dry, colonised hemp shives which can be reactivated using water and flour (Ecovative Design LLC 2014).

To prototype with SGM method, we choose to leverage the capacity of chlamydospores synthesis of *G. lucidum* and activate them by moisturisation. *G. lucidum* — also known as *linghzi*, *reishi*, for its long praised medicinal virtues (Blanchette 2020) — was also selected because of its widest array of carbohydrate-active enzymes (Zhou et al. 2018; Chen et al. 2012) ecologically competitive growth rate and conditions. Because fungal growth of model white rot species *Physisporinus vitreus* has been reported to increase with water activity (Fuhr et al. 2011), and enzymes catalytic activity are reportedly optimal at an activity of 0.99 ((Gervais & Molin 2003), we decided to use mineralised water in this study. As a complementary and comparative prototyping method, this study investigates the addition of a *G. lucidum* liquid culture inoculum in sterile beech wood particles. In all experiments, the colonisation at the forming stage was stopped after at least four days. A careful manual excavation was then performed, the grown mycelia were placed in a clean and moist environment, and ripened in the dark at 25 ºC for four days. After this, the specimens were dried at 60 ºC for 24h.

3.1 *Material inoculation and sterilization*

A millet-grown spawn of species *G. lucidum* (reference M9726) was acquired from Mycelia BVBA (Nevele, Belgium). The spawn was stored at a constant 4 ºC and 65 % RH. The substrate of the specimens is made of European beech wood *Fagus sylvatica* of a 0.75 - 3.0 mm granulation (Räuchergold type HB 750/2000, J. Rettenmaier & Söhne GmbH + Co KG, Rosenberg, Germany). The beech wood was moisturised at a 70% MC with mineralised water, not sterilised, and inoculated in a 16 wt% ratio with the *G. lucidum* spawn. The mix was incubated in a filtered 5L polypropylene box (SacO2, Deinze, Belgium) for 7 days at 25 ºC in the dark, then broken down manually and dried at 25 ºC for 24h. We confirmed the presence of chlamydospores with this cultivation technique by scanning electron microscopy (SEM), see Figure 1.

To prototype with liquid culture inoculation, dry European beech wood of previously described reference was sterilised for thirty minutes at 121 ºC, and a previously referenced strain of *G. lucidum* was prepared with a non-disclosed nutrient broth in liquid submerged fermentation.

Table 1. Classification of variables of morphological control.

Variable	Class	Stage of control	Modality of continuous control	Sterility	Resolution
Organic carbon	Endogenous	Setup, continuous	Liquid delivery	High	High
Carbon to nitrogen ratio	Endogenous	Setup, continuous	Liquid delivery	High	High
Ashes	Endogenous	Setup, continuous	Liquid delivery	High	High
Moisture content	Endogenous	Setup, continuous	Liquid delivery	Varying	High
Particles characteristics	Endogenous	Setup, continuous	–	Varying	High
Fungal strain	Endogenous	Setup, continuous	Liquid delivery	High	High
Temperature	Exogenous	Continuous	Radiating heat, liquid delivery	Varying	Medium
Light	Exogenous	Continuous	Radiation	–	Medium
Relative humidity	Exogenous	Continuous	Aerosol	–	Low
CO2 or O2	Exogenous	Continuous	Gaseous downward delivery	–	Low

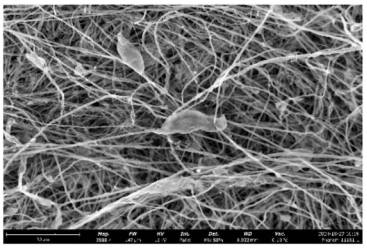


Figure 1. *Chlamydospores* present in cultivated *G. lucidum mycelia*.

3.2 *Monitoring and Evaluation*

While conventional 3D printing techniques do not involve living organisms, the integration of natural binders such as mycelium necessitates active monitoring of environmental conditions. Therefore, the print chamber has been equipped with temperature, humidity, and air quality sensors (BME680 Sensor), in addition to an RGB camera, which serve to track and evaluate the outcomes of the 3D printing process as shown in Figure 2.

4 RESULTS

Following three preliminary experiments in which water and bio-based hydrogel were applied to both inoculated and non-inoculated beech powder substrates to cultivate mycelium within the powder bed resulted in the growth of contaminant green and black molds. These outcomes were attributed to contamination within the liquid medium and substrate, along with exposure to open air. Additionally, the inoculated substrate exhibited water-repellent properties, impeding effective moisture absorption. To address these challenges, the design of the 3D printer was re-engineered and rebuilt to enable direct injection of liquid into the substrate under controlled laminar flow conditions. This re-engineering aimed to mitigate contamination risks and establish a more favorable environment for mycelial development.

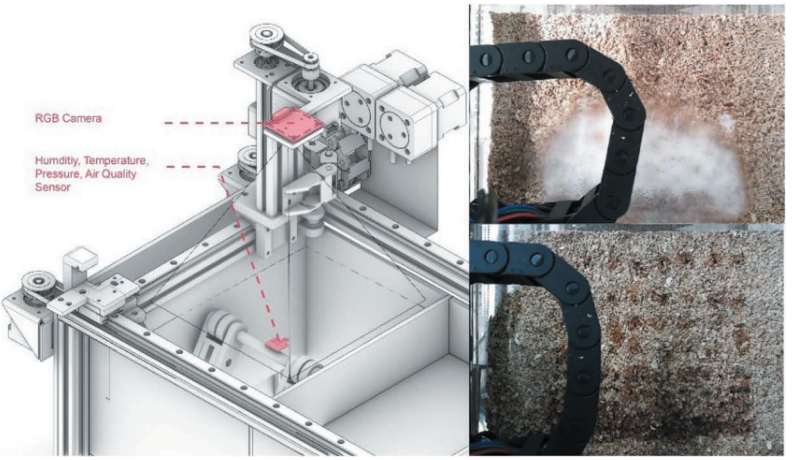


Figure 2. Left Sensor and Camera Position on 3D printer Right Photos taken by the camera in the first Experiment.

4.1 *Experiments*

In the first experiment, the objective is to understand injection point spacing impact on mycelium growth coverage and rate, therefore an inoculated substrate was injected with tap water at grid point distance of 0.5, 1, 1.5, 2,3 cm, ranging from a dense to a more widely spaced configuration. The results indicate rapid mycelial growth over a period of four days, particularly within point distances of 0.5 - 1 centimeter (see Figure 2 right).

In the second experiment, the objective is similar to experiment one however an uninoculated (raw) sterile substrate substitutes the inoculated substrate and instead of water, liquid culture inoculum is injected. The grid configuration is similar to experiment one; however, the results demonstrate a slower growth rate and significant surface contamination due to the substrate's lack of inoculation and therefore exposure to contamination during growing period. Black and green mold contamination was evident, which impacted the results and white fungi only appear in the grid point distance of 0.5 cm.

To refine our 3D printing parameters based on findings from the first two experiments, we perform a third experiment with the objective to investigate how variations in liquid volume and injection density affects growth. We investigate two parameters in this test. First parameter is differences in grid point distances as can be seen in right side of Figure 3 in first chamber, second parameter is differences in the amount of water injected inside the inoculated substrate ranging from 3, 1, 0.5 ml with the cubes made out of dense grid of 0.5 cm distance in second chamber. This setup, however, resulted in a high level of contamination both on the surface and within the 3D-printed structures. The contamination was due to residual inoculated liquid in the tubes connected to the nozzle from prior experiment. Results from first chamber test illustrate that cubes with point distances of 0.5 and 1 cm were connected to each other while the cube with 0,5 cm distance points on the outer boundary managed to stay separated see right side of Figure 3.

The outcomes of second chamber shows that 3 ml of water resulted in a cube however lower values didn't have any outcome.

Due to contamination in the third experiment similar test model was repeated in the fourth experiment, the objective of this test is to understand the impact of layering raw and inoculated beech powder in mycelium growth and coverage, therefore the substrate consisted of a blend of inoculated and uninoculated (raw) wood powder to enhance water absorption. Nevertheless, instead of homogeneously mixing the two powders, raw and inoculated beech powder was introduced in distinct layers in the chambers. The tab water was also mixed with wetting agent to increase the spreading and penetrating properties of water in substrate, the aim of this experiment was to increase water penetration and capacity of the substrate. The findings show that while the inoculated substrate became saturated with mycelium, the non-inoculated layers exhibited no

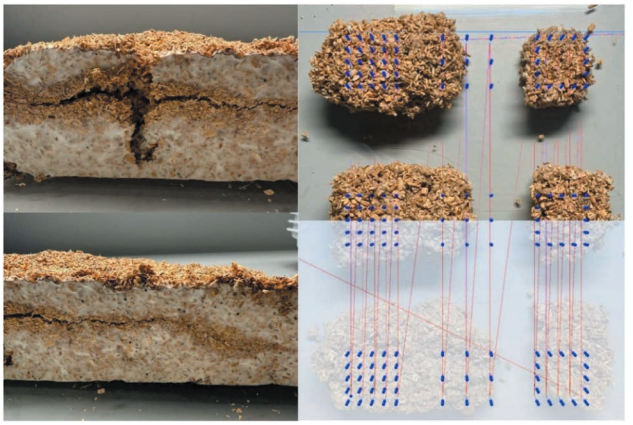


Figure 3. Left Fourth Experiment layered inoculation Right Third Experiment Inoculated Cubes.

mycelial growth due to their longer growth period, leading to a multi-layered object which could be separated from each other (see Figure 3 left).

In the fifth experiment, the aim is to increase the scale of 3D printed objects, thus a large box of size 10cm was 3D printed directly within the inoculated substrate using injection of 3ml tap water with single wall of points similar to the boundary grid of third experiment, however no growth was visible.

5 CONCLUSIONS

The experiments conducted highlight critical insights into the challenges and opportunities in cultivating mycelium using SGM method, particularly focusing on contamination control, water absorption, and mycelial growth patterns. By comparing the conducted experiments as shown in Table 2 the following strengths and weaknesses can be drawn.

Table 2. Experiment Details on Mushroom Growth under Different Conditions.

No.	Injected Liquid	Water Amount (ml)	Substrate	Test model	Growth Rate (days)	Duration (days)	Contamination	Results
1	Water	6	Inoculated Beech Poweder	Gradient of points	4	4	-	0.5 - 1 cm distance show full growth
2	liquid culture inoculum	6	Beech Powder	Gradient of Points	11	11	Surface – Green & Black mold	0.5 cm distance show average growth
3	Water	3, 1, 0.5	Inoculated Beech Poweder	Nine Boxes of Varying point density	4	11	Inside & Surface Black Mold	Poor growth due to contamination in tubes
4	Water + Wetting Agent	3, 1, 0.5	Raw + Inoculated Beech Poweder	Nine Boxes of Varying point density	4	11	Surface – Black Mold	Layered growth due to difference in layer mixture
5	Water	3	Inoculated Beech Poweder	One box	4	4	-	-

Strengths: Inoculated powder demonstrated a promising growth rate with grid point distances of 0.5–1 cm, achieving rapid mycelial proliferation within four days. it also demonstrates resistance to microbial competition during growth period resulting in less contamination in the process. The adjustment of substrate layering and incorporation of a wetting agent in the fourth experiment further enhanced water penetration, creating a gradient structure with distinct zones of mycelial growth which can be explored more in different ratios in future experiments. The presence of chlamydospores in the produced parts after drying process enables reusability of the 3D printed parts through grinding down of the manufactured objects for production of another object.

Weaknesses: The presence of contamination on top layer is evident in all experiments, the amount relies both on the level of sterility of conducted experiments and involved ingredients. The low viscosity of tap water is another point to consider which leads to more growth in the lower parts of the powder bed and short evaporation period which reduces growth period in the substrate.

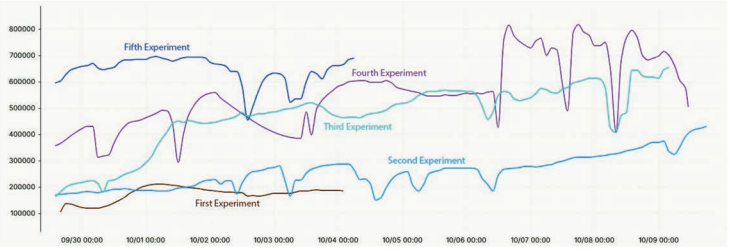


Figure 4. CO2 Measurement in experiments.

Carbon dioxide readings from the sensor reveals that by increasing the depth of chamber the amount of CO2 readings increase, as in the fifth experiment the maximum depth of 12 cm is used which leads to highest reading in comparison with first experiment which has lowest depth (see Figure 4). Experiments with Inoculated Substrate show period of increase to a peak indicating a high rate of mycelial activity and then decrease to flatten and experience multiple cycles, while inoculated liquid experiment suggest a dynamic environment with possible cycles of mycelial respiration and/or contamination.

By analyzing the results from experiments, it becomes evident that inoculated powder with enhanced water absorption and substrate layering strategy, currently offers a more robust solution for producing structured mycelial growth.

6 DISCUSSION

In the state-of-the-art, 3D printing strategies for mycelium materials have been employing the extrusion of an inoculated substrate, the injection of mycelium onto a 3D printed structure, or the direct exposition of a 3D printed structure to a grain spawn. While this approach is effective for producing homogeneous components, the SGM method that we introduce in this study opens the design space towards creating heterogeneous structures with differentiated morphologies. This is achieved through controlled growth originating from variations in point density and powder type. Additionally, SGM has the notable advantage of allowing the cultivation of parts without the need for support structures. In addition to that, 3D printing with chlamydospores still present and active after drying allows grinding and remanufacturing and reusability of the 3D printed parts. The utilisation of this novel 3D printing technique combined with the deployment of a monitoring system designed for the fine events of biological growth heralds a transformative approach that enhances developmental outcomes. This method facilitates the study of mycelium growth and morphological differentiation under a variety of environmental conditions.

Materials and methods potential improvements for future studies:

– Containment and Cleanability: Clean Chamber with Filtered Ventilation and UV sterlisiation before 3D printing can omit the use of laminar flow.
– Added Technologies: Powder Deposition for multilayered powder experiments, Multimaterial Injection for injecting different liquids in the powder.

– Environmental control: light, gas, temperature, humidity control inside the chambers can turn chambers into bioreactor for controlling growth
– Injected liquid: Gel matrix, Catalyst, Preventive Liquids, Multiple species of mycelium and bacteria to test interaction of species, less evaporation, faster growth and stopping growth to control the outcomes.

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